

EduFlow — International Student Analytics Platform

The problem

Universities enroll thousands of international students every year. Behind each one is a long, fragile journey: applications, document verification, visa processing, enrollment, term-by-term registration, and eventually graduation or an early exit. That entire journey is tracked across disconnected systems — a student information system here, a visa management tool there, spreadsheets filling the gaps.

The result: nobody in the institution has a clear, live picture of what's actually happening across their international student population. Answering a question like *"what's our visa approval rate this year compared to last?"* or *"which programs have the highest mid-program dropout rates among international students?"* requires manual data pulls, hours of cleaning spreadsheets, and by the time an answer is ready, it's already outdated.

This is a solved problem in e-commerce, finance, and logistics. It is not solved in international higher education.

The opportunity

1.1 million international students studied in the US in 2023/24. Every university enrolling them is legally required to track their status (SEVIS compliance), academically motivated to retain them (tuition revenue, rankings), and institutionally invested in their outcomes (accreditation, reputation).

None of them have a real-time, unified analytics layer sitting on top of all that data.

What EduFlow does

EduFlow is a data platform that captures the international student lifecycle as a stream of events — every application submitted, document uploaded, visa decision received, course registered — and turns that stream into live, actionable analytics for the people who need it.

For the international student office:

- Real-time enrollment funnel: where are students dropping off, and at which stage?
- Visa processing times by country and consulate
- Document compliance rates and bottlenecks
- Early warning indicators for students at risk of withdrawal

For academic departments:

- Enrollment trends by program, school, and country of origin
- Retention rates broken down by degree level and funding source

- Term-by-term registration patterns

****For university leadership:****

- Year-over-year enrollment trends
- Tuition revenue forecasting based on pipeline conversion rates
- Compliance reporting (SEVIS, accreditation) without the manual effort

Who this is for

****Primary users (the people who sit in front of the dashboard every day):****

International student advisors, DSOs (Designated School Officials), and registrars. People who currently do this work manually, in spreadsheets and SIS exports, and who would immediately recognize the value of having it automated and live.

****Secondary users (the people who ask questions and read reports):****

Deans, provosts, enrollment management leadership. They don't need the operational detail; they need the trend lines and the forecasts.

****The buyer:****

The university itself, specifically the Office of International Affairs or Enrollment Management. This is a B2B institutional sale, not a consumer product. Universities already spend significantly on SIS licenses, compliance tools, and BI software; EduFlow fits naturally into that budget category.

How it's built

The platform is built on the same infrastructure patterns used by companies like Airbnb, Uber, and LinkedIn to handle high-volume, time-sensitive data:

- ****Event streaming (Kafka):**** every student action is captured the moment it happens, in real time, not in a nightly batch export
- ****Orchestrated processing (Airflow):**** automated pipelines validate, clean, and transform the raw events on a reliable schedule, with alerts when something breaks
- ****Distributed transformation (Spark):**** handles the scale of a large university's entire international student population without manual effort
- ****Cloud data warehouse (Redshift/S3):**** all cleaned, curated data lives in a structured warehouse, queryable instantly
- ****Live dashboard:**** a clean, purpose-built interface that surfaces the metrics that matter to each user type, without requiring them to write a single query

The entire infrastructure is containerized (Docker/Kubernetes) and provisioned as code (Terraform), meaning it can be deployed to a new university environment in hours, not weeks.

What's next

The immediate goal is a working prototype demonstrating the full pipeline end-to-end: event ingestion → automated processing → live dashboard, running against a realistic synthetic dataset modeled on a real university's international student population. That prototype is the proof-of-concept for the first university pilot conversation.